CAPSTONE PROJECT PROPOSAL TITLE

A Capstone Project and Research Study Proposal

Presented to the

Department of Information Technology
University of Science and Technology of Southern Philippines
Oroquieta City

In Partial Fulfillment of the Requirements for
Capstone Project and Research Study Leading to the Degree in
Bachelor of Science in Information Technology

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APPROVAL SHEET

(Please attach here a scanned copy of the approved DPM Form: FM – USTP – ACAD – 10)

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

The Background of the Study explains the current situation related to the problem being addressed. Begin by describing the general environment where the problem exists, such as an organization, institution, community, or industry. The discussion should gradually narrow down from the broader context to the specific issue that motivated the development of the proposed system.

Describe here existing processes, workflows, or systems currently used by the organization or stakeholders. If the current system is manual or inefficient, the discussion should clearly highlight the limitations or challenges encountered by users. The purpose of this section is to demonstrate that the problem is real and significant.

The background should end with a clear statement emphasizing the need for a technological solution. This naturally leads to the formulation of the problem statement.

1.2 Statement of the Problem

This section presents the specific problems that the study aims to address. The problems should be written as clear and concise research questions that the system will attempt to solve. The problem statement must be derived directly from the issues discussed in the background of the study.

Typically, the problem statement begins with a general problem followed by several specific questions. These questions should focus on the development, functionality, and effectiveness of the proposed system.

The problem statement should guide the entire research. Each question raised in this section must eventually be answered through the system implementation and evaluation described in later chapters.

1.3 Statement of Objectives

The objectives describe what the study intends to accomplish. This section usually includes a general objective followed by several specific objectives.

The general objective describes the overall purpose of the project, which is typically the development of an information system or technological solution. The specific objectives provide more detailed goals that support the general objective.

These objectives should be achievable and directly related to the problems stated earlier. Each objective should describe a measurable outcome, such as designing a system component, implementing a module, or improving a specific process.

1.4 Significance of the Study

The Significance of the Study explains the importance of the project and identifies the individuals or groups who will benefit from it. Students should describe how the proposed system will improve existing processes or provide new capabilities for its users.

Possible beneficiaries may include the client organization, employees, administrators, researchers, and future developers. The discussion should explain how each stakeholder group benefits from the proposed system.

This section demonstrates the value and relevance of the project beyond academic requirements.

1.5 Scope and Limitation of the Study

The Scope and Limitations section defines the boundaries of the project. The scope explains what the system will include, such as the modules, features, and user roles that will be implemented.

On the other hand, the limitations identify aspects that are outside the coverage of the study. These limitations may be due to time constraints, technological restrictions, or lack of available resources.

Clearly defining the scope and limitations helps set realistic expectations for the system and prevents misunderstanding about what the project intends to deliver.

1.6 Definition of Terms

This section provides operational definitions for technical terms used in the manuscript. The purpose is to ensure that readers clearly understand specialized terms, acronyms, or concepts mentioned throughout the study.

Definitions should be written in alphabetical order and presented in a concise manner. Each term should be defined according to how it is used within the context of the study.

CHAPTER II

REVIEW OF RELATED LITERATURES

2. 1 Review of Literature

The Review of Literature discusses scholarly studies, theories, and technological developments relevant to the project. You consult reliable sources such as academic journals, conference papers, books, and credible online publications.

The literature review should not simply summarize sources. Instead, it should analyze how previous studies relate to the proposed system. Students should highlight important findings, methodologies, and insights that support the design and development of their system.

The discussion should also identify limitations of previous research and how the current project addresses these gaps.

2.2 Review of Related Application

This section focuses on existing systems or software applications that perform similar functions to the proposed system. Each system should be briefly described, including its purpose, features, and limitations.

A comparison table summarizing these systems should be presented, followed by a narrative explanation of their similarities and differences with the proposed project.

2.3 Synthesis and Gap Analysis

The synthesis integrates the insights gathered from the literature review and related applications. Instead of describing each source separately, this section summarizes the overall trends and findings.

You should identify patterns, common strengths, and recurring limitations among existing studies and systems. The synthesis should conclude by explaining the specific gap that the proposed system intends to address.

This section justifies the relevance and necessity of the project.

CHAPTER III

METHODOLOGY

3.1 Research Methodology

The research methodology describes the development approach used in the project. Students should explain the development model adopted, such as Agile, Waterfall, Rapid Application Development, or other appropriate methodologies.

The discussion should explain the phases of the methodology and how each phase contributes to the development of the system. This provides a structured framework for the project implementation.

3.2 Data Gathering Procedure

This section explains how needed information to build your system is being collected. Students should describe the methods used to understand the problem and gather system requirements.

Common things to discuss includes Sources of data (users, stakeholders, documents), Methods used (interview, survey, observation) and Purpose of data gathering.

The discussion should explain how these methods helped identify system requirements and guide the development of the proposed solution.

3.2.1 Research Instrument

This describes the tools used to collect data. This includes Questionnaires (if survey), Interview Guide (if interview) and observation checklist.

Explain what instrument you used, what it contains, and what type of data it collects.

3.2.2 Sampling Method

This explains who participated in your data gathering and how they were selected. Discussion includes Type of sampling method (purposive, random, etc.), Number of respondents and criteria

3.2.3 Statistical Treatment of Data

This explains how you analyzed the data collected, Tools Used (percentage, mean, Likert Scale), and purpose of analysis.

Discuss how responses were interpreted.

3.3 Description of the Current System (Information Gathering)

This describes the existing process/system before your solution. Discuss here the current workflow, tools currently used and problems encountered. Explain the current situation clearly, then highlight its limitations.

3.4 System Architecture and Development

This explains how your system is structured and developed.

3.4.1 Overview of the Proposed System

Discuss here the general description of the system, its main features and the users. Provide a high-level explanation only (no deep technical details yet).

3.4.2 System Objectives

Discuss here what the system aims to achieve. Align this with your Chapter 1 objectives but focus on system functionality.

3.4.3 System Tools and Requirements

This explains what technologies are used to build the system. Explain why each tool is used.

Programming Languages

- *Languages used (e.g., JavaScript, Python)*
- *Purpose of each*

Hardware Requirements

- *Devices needed (PC, server, camera, etc.)*

Software Requirements

- *Tools and frameworks (e.g., React, Firebase)*

3.4.4 System Flow/Functions

This explains how the system works step-by-step. Discuss here the Input-Process-Output and User Interactions. Basically, narrate the flow of activities inside the system.

3.4.5 Scope and Limitations of the System

Discuss what the system can do and cannot do. Be specific and realistic.

3.4.6 Calendar of Activities

This shows your project timeline. Present a Gantt Chart that shows your activities from planning to deployment. Provide a detailed discussion of the chart.

3.5 Conceptual Design and Implementation

3.5.1 Database Design

Present your database design and explain how data is structured. (can either be ERD or NOSql Collection)

3.5.2 System Design

This presents the visual representation of your system.

Use Case Diagram

Shows users and system interactions

Context Diagram

Shows system boundaries and external entities

Data Flow Diagram

Shows data movement within the system